

Mortgage Tax Document Processing Architecture

1. Purpose

This document outlines the architectural approach for an enterprise mortgage document processing system focused on tax document classification, extraction, and MGIC income calculation. The objective is to minimize manual interpretation, automate extraction and calculations, and maintain a human-in-the-loop validation workflow.

2. Design Principles

- Hybrid architecture combining .NET orchestration with Python AI services
- Use Azure Document Intelligence where structured tax forms are stable
- Use Python extraction for complex business tax forms with layout variability
- Classifier-first architecture: classify → route → extract → validate → calculate
- Human-in-the-loop validation for all extracted financial values
- Confidence-based routing for review workflows
- Maintain full audit trail of extracted values and corrections

3. High Level Architecture

Upload → Document Classification → Extraction Engine → Normalization → MGIC Calculation Engine → Human Review → Export

4. Technology Stack Strategy

- .NET Backend: orchestration, APIs, workflow management
- Python Microservices: document extraction, ML models, financial calculation engine
- Azure Document Intelligence: structured tax form extraction
- Azure Service Bus: event-driven processing pipeline
- Azure Blob Storage: document storage
- SQL Server: metadata and extracted field storage
- React Frontend: document review and correction interface

5. Extraction Strategy

The system uses a routing model where each classified tax form is processed by the optimal extraction engine depending on layout stability and Azure model support.

6. Form Processing Strategy

Tax Form	Use Azure Prebuilt Model	Use Python Extraction	Reason
W-2	Yes	No	Stable structure and supported by Azure DI
1099-NEC / 1099-R / 1099-MISC	Yes	No	Consistent tax form layouts
Form 1040	Yes	No	Primary personal tax return supported by Azure
Schedule B	Yes	No	Supported prebuilt Azure model
Schedule C	Yes	No	Well structured, supported by Azure
Schedule D	Yes	No	Supported by Azure tax models
Schedule E	Yes	No	Rental income extraction supported
Schedule F	Yes	No	Farm income layout consistent
Schedule K-1 (Partnership)	No	Yes	Complex multi-entity layout not supported by Azure
Schedule K-1 (S-Corp)	No	Yes	Ownership and distribution fields require custom logic
Form 1065	No	Yes	Large multi-page business tax return
Form 1120-S	No	Yes	Business income analysis requires custom logic
Form 1120	No	Yes	Corporate tax return with variable structure

7. Python Extraction Technology Stack

- PaddleOCR / Tesseract for OCR processing
- pdfplumber and PyMuPDF for PDF parsing
- LayoutParser for layout detection
- Camelot / Tabula for table extraction
- FastAPI for extraction microservices
- Pandas / NumPy for financial calculation logic

8. Calculation Engine

Financial calculations such as qualifying income, ownership adjustments, and depreciation add-backs should be implemented in a deterministic Python calculation engine rather than AI models. This ensures auditability and compliance with mortgage underwriting guidelines.

9. Human-in-the-Loop Workflow

All extracted data is presented in a review interface where an underwriter can verify values against the source document. Corrections are captured and fed back into the training dataset to improve future extraction accuracy.

10. Key Benefits of Hybrid Approach

- Reduced Azure AI processing costs
- Improved scalability for large document volumes
- Greater flexibility for complex business tax forms
- Ability to evolve extraction logic independently from Azure services
- Better auditability for financial calculations

2020 W-2 and EARNINGS SUMMARY



Employee Reference Copy		2020	
W-2		Wage and Tax Statement	
OMB No. 1545-0008			
Copy C for employee's records.			
d Control number	Dept.	Corp.	Employer use only
000050	R3/TKW		
c Employer's name, address, and ZIP code			
RNR IT SOLUTIONS INC 7800 PRESTON ROAD SUITE 126 PLANO, TX 75024 3240			
Batch #92639			
e/f Employee's name, address, and ZIP code			
SANTOSH KALE 4007 LAKE UNION HILLWAY ALPHARETTA, GA 30004			
b Employer's FED ID number	a Employee's SSA number		
47-1550133	XXX-XX-2052		
1 Wages, tips, other comp.	2 Federal income tax withheld		
19666.00	726.08		
3 Social security wages	4 Social security tax withheld		
19666.00	1219.29		
5 Medicare wages and tips	6 Medicare tax withheld		
19666.00	285.16		
7 Social security tips	8 Allocated tips		
9	10 Dependent care benefits		
11 Nonqualified plans	12a See instructions for box 12		
14 Other	12b		
	12c		
	12d		
	13 Stat emp Ret. plan 3rd party sick pay		
15 State	Employer's state ID no.	16 State wages, tips, etc.	
GA	3358911-OV	19666.00	
17 State income tax	18 Local wages, tips, etc.		
985.79			
19 Local income tax	20 Locality name		

This blue section is your Earnings Summary which provides more detailed information on the generation of your W-2 statement. The reverse side includes instructions and other general information.

1. Your Gross Pay was adjusted as follows to produce your W-2 Statement.

	Wages, Tips, other Compensation Box 1 of W-2	Social Security Wages Box 3 of W-2	Medicare Wages Box 5 of W-2	GA. State Wages, Tips, Etc. Box 16 of W-2
Gross Pay	19,666.00	19,666.00	19,666.00	19,666.00
Reported W-2 Wages	19,666.00	19,666.00	19,666.00	19,666.00

2. Employee Name and Address.

SANTOSH KALE
4007 LAKE UNION HILLWAY
ALPHARETTA, GA 30004

© 2020 ADP, Inc.

1 Wages, tips, other comp.	2 Federal income tax withheld
19666.00	726.08
3 Social security wages	4 Social security tax withheld
19666.00	1219.29
5 Medicare wages and tips	6 Medicare tax withheld
19666.00	285.16
d Control number	Dept.
000050	R3/TKW
c Employer's name, address, and ZIP code	
RNR IT SOLUTIONS INC 7800 PRESTON ROAD SUITE 126 PLANO, TX 75024 3240	
b Employer's FED ID number	a Employee's SSA number
47-1550133	XXX-XX-2052
7 Social security tips	8 Allocated tips
9	10 Dependent care benefits
11 Nonqualified plans	12a See instructions for box 12
14 Other	12b
	12c
	12d
	13 Stat emp Ret. plan 3rd party sick pay
e/f Employee's name, address and ZIP code	
SANTOSH KALE 4007 LAKE UNION HILLWAY ALPHARETTA, GA 30004	
15 State	Employer's state ID no.
GA	3358911-OV
17 State income tax	18 Local wages, tips, etc.
985.79	
19 Local income tax	20 Locality name
Federal Filing Copy	
W-2	
Wage and Tax Statement	
OMB No. 1545-0008	
Copy B to be filed with employee's Federal Income Tax Return.	

1 Wages, tips, other comp.	2 Federal income tax withheld
19666.00	726.08
3 Social security wages	4 Social security tax withheld
19666.00	1219.29
5 Medicare wages and tips	6 Medicare tax withheld
19666.00	285.16
d Control number	Dept.
000050	R3/TKW
c Employer's name, address, and ZIP code	
RNR IT SOLUTIONS INC 7800 PRESTON ROAD SUITE 126 PLANO, TX 75024 3240	
b Employer's FED ID number	a Employee's SSA number
47-1550133	XXX-XX-2052
7 Social security tips	8 Allocated tips
9	10 Dependent care benefits
11 Nonqualified plans	12a
14 Other	12b
	12c
	12d
	13 Stat emp Ret. plan 3rd party sick pay
e/f Employee's name, address and ZIP code	
SANTOSH KALE 4007 LAKE UNION HILLWAY ALPHARETTA, GA 30004	
15 State	Employer's state ID no.
GA	3358911-OV
17 State income tax	18 Local wages, tips, etc.
985.79	
19 Local income tax	20 Locality name
GA. State Reference Copy	
W-2	
Wage and Tax Statement	
OMB No. 1545-0008	
Copy 2 to be filed with employee's State Income Tax Return.	

1 Wages, tips, other comp.	2 Federal income tax withheld
19666.00	726.08
3 Social security wages	4 Social security tax withheld
19666.00	1219.29
5 Medicare wages and tips	6 Medicare tax withheld
19666.00	285.16
d Control number	Dept.
000050	R3/TKW
c Employer's name, address, and ZIP code	
RNR IT SOLUTIONS INC 7800 PRESTON ROAD SUITE 126 PLANO, TX 75024 3240	
b Employer's FED ID number	a Employee's SSA number
47-1550133	XXX-XX-2052
7 Social security tips	8 Allocated tips
9	10 Dependent care benefits
11 Nonqualified plans	12a
14 Other	12b
	12c
	12d
	13 Stat emp Ret. plan 3rd party sick pay
e/f Employee's name, address and ZIP code	
SANTOSH KALE 4007 LAKE UNION HILLWAY ALPHARETTA, GA 30004	
15 State	Employer's state ID no.
GA	3358911-OV
17 State income tax	18 Local wages, tips, etc.
985.79	
19 Local income tax	20 Locality name
GA. State Filing Copy	
W-2	
Wage and Tax Statement	
OMB No. 1545-0008	
Copy 2 to be filed with employee's State Income Tax Return.	